Molecules & Compounds

Exam code: 0620 & 0971

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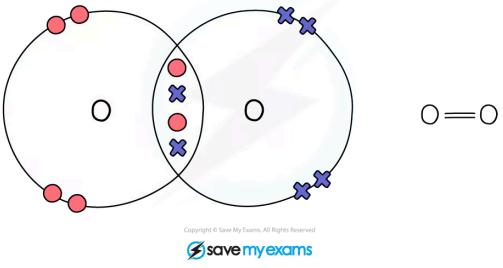
Covalent bonds in complex covalent molecules

Extended Tier Only

- Some atoms need to share more than one pair of electrons to gain a full outer shell of electrons
- If two adjacent atoms share two pairs of electrons, two covalent bonds are formed, also known as a **double bond**
- If two adjacent atoms share three pairs of electrons, three covalent bonds are formed, also known as a **triple bond**

Oxygen:

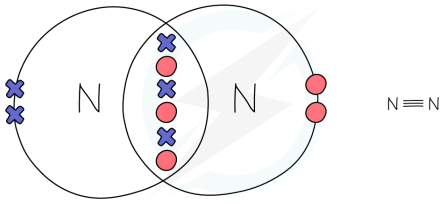
When 2 oxygen atoms react, they share 2 pairs of electrons to form a double bond



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

electrons to form a triple bond



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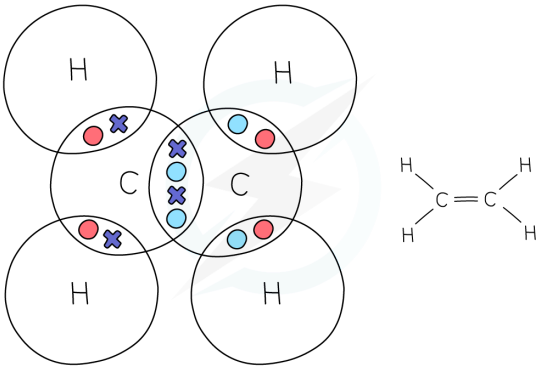
Ethene:

- In ethene, the 2 carbon atoms share 2 pairs of electrons
- This is known as a double bond



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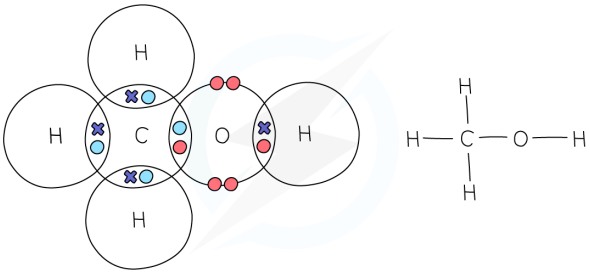
Methanol:



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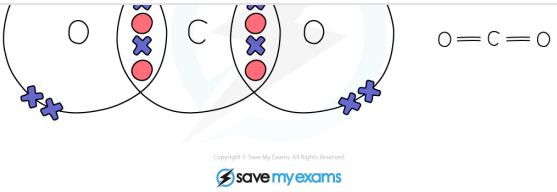
Carbon Dioxide:



Examiner Tips and Tricks

Be careful when drawing dot-and-cross diagrams, it is a common mistake for students to draw the wrong type of diagram.

Remember, if the compound contains **metal** and **non-metal**, it is an **ionic** compound and you need to draw the ions separated, with square brackets around each ion, together with a charge.

If the compound contains **non-metal** atoms only, it is a **covalent** compound, the shells should overlap and contain one or more pairs of electrons.

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